
MUSTAFA HIJAZI

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PROFESSIONAL SUMMARY

Mechanical Engineer with experience in MEP, CAD modelling, power generation, prototyping, and robotics. Strong problem solver with a passion for learning and delivering results in a collaborative environment.

EDUCATION

Master of Science in Mechanical Engineering - Automotive and Motorsport Engineering Track

Politecnico di Milano – Milan, Italy

2026 - Present

Bachelor of Engineering (BE): Mechanical Engineering

2026

American University of Beirut – Beirut, Lebanon

WORK HISTORY

Engineering Intern

08/2025

Jubaili Bros – Saida, Lebanon

- Learned generator components, wiring schemes, and electrical system layouts.
- Assembled control panels and power generators with the help of experienced technicians
- Viewed the design and fabrication of chassis and cases that house these generators

Mechanical Engineering Intern

06/2025 - 07/2025

Alef Group – Sharjah, UAE

- Reviewed HVAC, drainage, fire protection, natural gas, and water layouts for 3 residential projects
- Drew HVAC layouts for two villas using AutoCAD
- Visited construction sites to observe and verify MEP services installation inspections

PROJECTS

Tri-copter with 360° Omnidirectional Maneuvering – Final Year Project

08/2025 – 11/2025

- Designed a custom UAV with a carbon fiber frame and 3D printed PETG arm mounts, focusing on strength and weight reduction under the guidance of Prof. Bilal Kaddouh
- Exposed the frame to 9G loading and verified its maximum stress is well below the failure stress
- Created CAD models, developed the wiring layout, and built a functional prototype

Human Robotic Interaction – PhD Thesis

08/2025 – 11/2025

- Assisted a PhD student on researching humanlike trajectories using minimum-jerk optimization guided by Prof. Daniel Asmar
- Successfully simulated a smooth trajectory for a 6-DOF robotic arm on Gazebo using ROS2 and Python

Autonomous Firefighting UAV

09/2025 – 11/2025

- Developed logic for the scooping-phase control loop of an autonomous firefighting aircraft based on the CL-415
- Drafted the control architecture, block, and state diagrams for the UAV under the guidance of Prof. Bilal Kaddouh

Robot Dog Locomotion in Complex Environments

02/2025 - 04/2025

- Worked with a team under Prof. Hussein Hussein to improve the locomotion capabilities of the Unitree Go1 dog robot in complex, unstructured environments.
- Tested High-level command, then moved on to Low-level control adjustments

Bicycle-Based Power Generator, Mechanical Design I Course

12/2024

- Designed a stationary pedaling machine that generates electrical power and stores it in a battery
- Built a working prototype from spare bicycle parts

Rotating Solar Panel, Process Instrumentation Course

05/2024

- Built a solar-tracking system using Arduino, LDR sensors, and a DC motor.
- Improved power efficiency by ensuring continuous sun alignment

SKILLS

- Software: SolidWorks, AutoCAD, Creo Parametric, MATLAB, Arduino, Ansys Fluent, ROS2, Gazebo
- Technical Skills: CAD modelling, FEA, Robotics, UAV design, Prototyping
- Other: Microsoft Office, Ubuntu

COMMUNITY SERVICE

Badiri NGO, Charity, Saida, Lebanon,

09/2024 – 10/2024

- Helped organize and prepare food for distribution

Extracurriculars

NASA Space Apps Competition - Beirut

10/2025

- Led a team to develop a rapid-turnaround solution for NASA's Terra Data Challenge

AFFILIATIONS

- American Society of Mechanical Engineers AUB Student Chapter, Member 2022 - Present

Licenses and Certifications

- [Data Fundamentals](#) – IBM SkillsBuild 07/08/2026
- [Data Classification](#) – IBM SkillsBuild 03/09/2026